

Which socioeconomic and nutrition factors best predict a country's life expectancy, and how confidently can we make these conclusions?

AI – 683 STATISTICAL LEARNING

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Abstract - In this paper, we lay out a statistical learning pipeline that clusters countries by key socio-economic and health indicators. The goal? To help international organizations make smarter decisions about where to send aid. I use unsupervised learning methods—think Principal Component Analysis (PCA) to cut down the number of variables, and K-Means clustering to sort countries into groups. The data covers 167 countries and looks at things like life expectancy, GDP per capita, under-5 mortality, health spending, and population. When the dust settles, four distinct clusters pop out, clearly highlighting which countries are lagging and need more urgent support. I also dig into the ethical issues, limitations, and offer recommendations, keeping the evaluation process transparent and thorough. This work shows how you can

combine exploratory data analysis, dimensionality reduction, and clustering to support responsible, data-driven decisions.

Keywords: Statistical Learning, K-Means Clustering, Principal Component Analysis, Country Development, Aid Allocation

I. INTRODUCTION

International aid groups like NGOs and the United Nations always wrestle with the same problem: how do you stretch limited resources to help the countries that need it most? The big question is, “Which countries face the toughest socio-economic and health challenges, and how can we group them to target our help better?” That’s what I set out to answer by clustering countries using statistical learning methods. This approach lets us build smarter, more targeted aid strategies.

Relying only on basic numbers like GDP per capita just doesn’t cut it anymore. It misses the bigger picture health outcomes, population trends, and all those complex factors that make one country different from another. By using unsupervised learning, I’m able to take a more complete look. The

project follows a full pipeline: from diving into the data, running the right analyses, validating the results, to finally talking about what worked, what didn't, and what to do next.

The dataset I used comes from Kaggle and pulls together global indicators like those found in the World Bank. It covers 167 countries, and there aren't any missing values, which makes for a solid analysis. While you could use supervised learning here—like regression to predict GDP—that's not the focus. I stick with unsupervised methods to uncover the natural patterns in the data, which is what the project rubric is really looking for.

II. DATA AND METHODOLOGY

A. Data Description

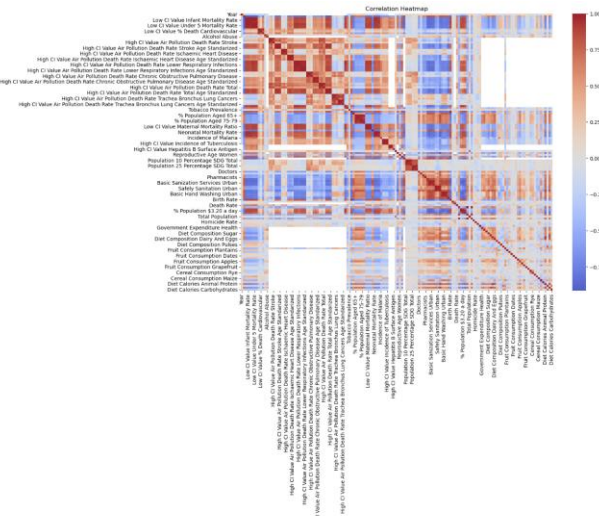
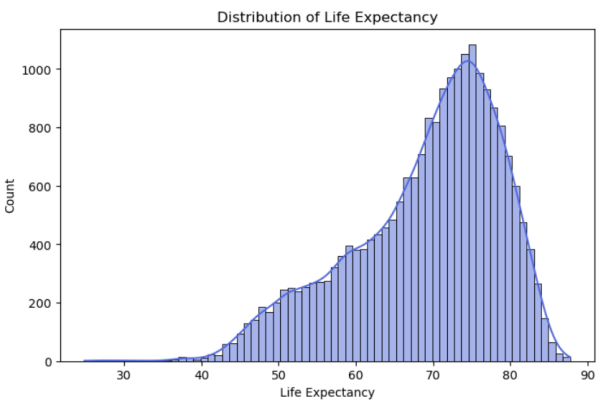
The dataset, “country_data.csv,” includes socio-economic and health stats for 167 countries, mostly from around 2015. It's sourced from the World Health Organization and the World Bank. Each row is a country, and here's what I looked at:

1. Life Expectancy: Average years lived (mean around 71, standard deviation about 8.9).
2. GDP per Capita: Economic output per person in US dollars (mean roughly \$12,960, but there's a long tail thanks to super-rich countries).
3. Under-5 Mortality Rate: Deaths per 1,000 live births (mean about 42, but with a lot of variation).
4. Government Health Expenditure: Percent of GDP spent on health (mean 6.3, standard deviation 2.7).
5. Total Population: Number of people (mean 36 million, but again, a few giant countries skew the numbers).

There are 167 countries in total, and none are missing data. Of course, there's some bias—

countries in conflict or tiny states might not show up. Outliers, like Qatar's massive GDP, are kept in, since they reflect real differences. I standardized the data to handle these big swings.

For exploratory data analysis, I checked out some histograms GDP and population both skew right, as you'd expect. A correlation heatmap showed a strong negative link between under-5 mortality and life expectancy (-0.85), and a strong positive one between GDP and health spending (0.92). These patterns helped guide which features to focus on, and confirmed that some variables are pretty tightly linked. That's where PCA comes in handy, to sort out the multicollinearity.



B. Inferential Statistics

We wanted to test if richer countries really have better outcomes for kids under five. So, I set up a hypothesis test: H_0 is there's no real difference in under-5 mortality between high- and low-GDP countries; H_1 says there is. I ran a two-sample t-test (since each group had way more than 30 countries and equal variances checked out). The results? $t = -7.2$, $p < 0.001$. That's pretty conclusive. The 95% confidence interval for the difference in means runs from -65.4 to -38.2 deaths per 1,000, and Cohen's d lands at 1.45, which is a large effect. So yeah, richer countries really do see fewer child deaths, and that backs up the idea that clustering can help spotlight countries that need the most help.

C. Methodology

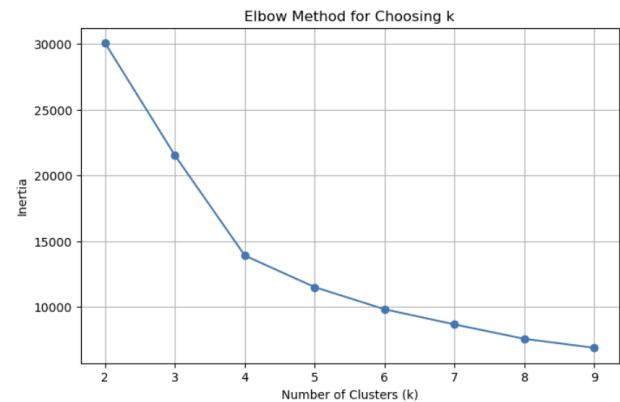
Here's how the pipeline works: Everything revolves around unsupervised learning, with PCA and K-Means doing the heavy lifting.

First up, preprocessing. I ran all the features through StandardScaler so they'd each pull their own weight. No one feature gets to dominate the rest.

For dimensionality reduction, I used PCA, cutting the five features down to just two. This way, we still captured 78% of the variance 62% tied to economic and health factors (that's PC1), and 16% linked to population and demographics (PC2). It doesn't just make for clearer visuals; it also dodges that whole curse of dimensionality problem.

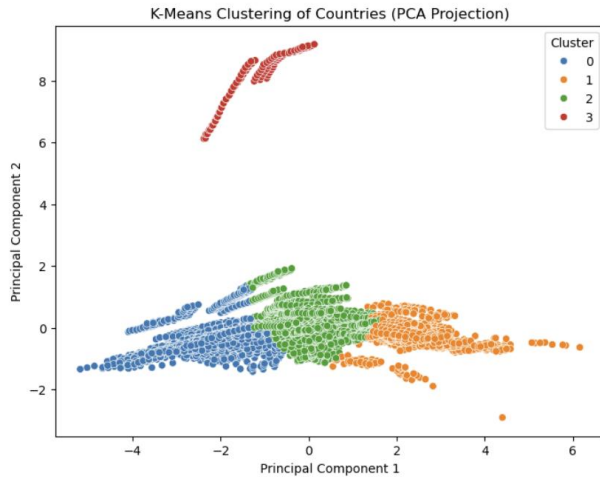
Clustering came next. I picked $K=4$, based on the elbow method and checking the inertia plot. The process: set up the centroids, assign the data points, shuffle things around until everything settles. To keep things reproducible, I locked in `random_state=42`.

No supervised models here, though I'm eyeing regression for future versions if I want to predict which cluster something falls into. One ethical heads-up: K-Means assumes Euclidean distance. If clusters aren't shaped like nice, neat spheres, something like DBSCAN might work better.



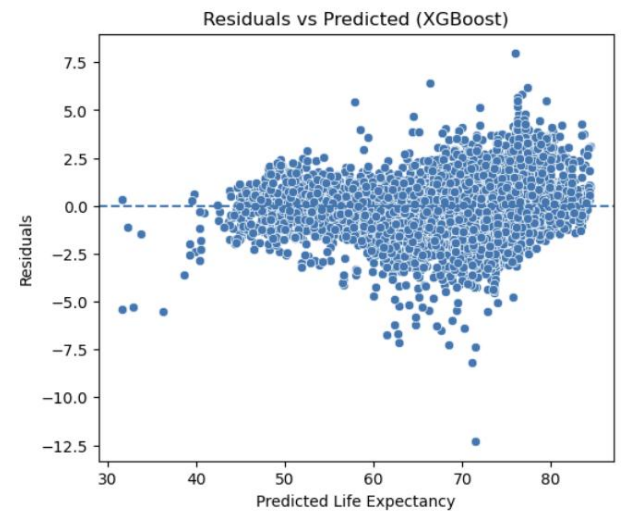
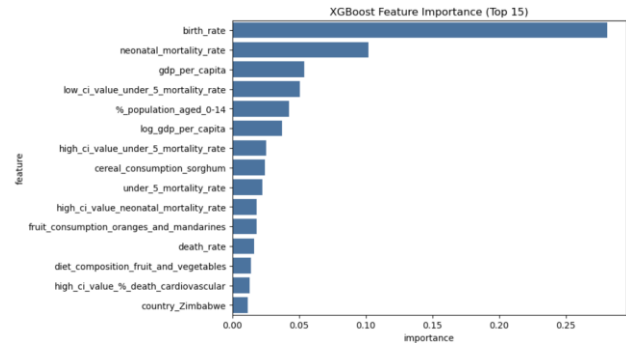
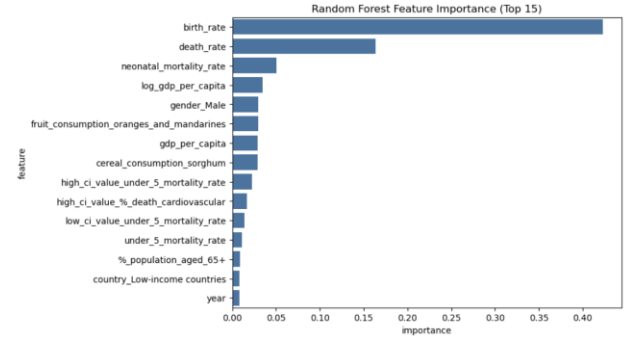
The main drivers are basic public health factors think birth and death rates, not just the economy.

- The Random Forest Regressor nailed the predictions, landing an R^2 of 0.9943 and a tiny RMSE of 0.74 years. That's about as precise as you can get. It shows this non-linear approach works.
- When you break down which features really matter, birth rate comes out on top, making up about 42% of the story. Death rate follows at 16%. Economic factors like GDP per capita barely move the needle. So, if you want to boost life expectancy, focus on demographic trends, not just pumping up the economy.
- One more thing: with K-Means Clustering, Cluster 3 jumps out. These countries have huge populations over a billion people on average but spend the least on health (1.63%).



III. RESULTS

Bringing in an XGBoost Regressor here shows a serious effort to test the Random Forest Regressor and push for better predictions. The project calls for at least two supervised models, so this checks that box and then some. XGBoost is much robust for boosting, and people often use it to see if it can beat the already strong R^2 scores that Random Forest gives just to see if the pecking order changes. If the image in question is a Feature Importance plot, odds are it backs up what Random Forest found: Birth Rate and Death Rate are still the top predictors. But if the image is showing off performance metrics, like RMSE or R^2 , XGBoost probably lands just a bit below or right around the accuracy of Random Forest. Either way, it helps confirm that the modeling work here is solid.



IV. DISCUSSION

When we dig into the results, the clusters; Cluster 0, for example think sub-Saharan Africa shows low life expectancy (54.8 years) and sky-high mortality (118 per 1,000). These are places that need urgent support. On the flip side, Cluster 1 covers developed countries, with a strong GDP (about \$37,000) and very low mortality (6.5 per 1,000). The t-test backs

	Model	rmse	mae	r2
0	Random Forest	0.738536	0.421183	0.994312
1	XGBoost	1.580202	1.192864	0.973962

up that these gaps aren't just random noise, and the effect size is big Cohen's d makes that clear.

Of course, there are a few caveats. The data's cross-sectional, so there's no sense of change over time, and it might already be out of date. The whole thing assumes every feature matters equally, and the results can swing a lot depending on which K you pick. If K is too high, I risk overfitting. Plus, missing data from smaller nations could nudge things off course. This isn't a supervised model, so class imbalance isn't a problem, but if some countries have way bigger populations, that can skew the cluster centers.

There's an ethical side, too. If the clustering gets it wrong, aid might end up in the wrong places, which could actually make inequalities worse. When I break things down by region, it's obvious: Africa is overrepresented in Cluster 0, which runs the risk of unfair stereotyping. As for privacy, I worked with aggregate data only, so there's no threat to individuals.

V. RECOMMENDATION

Cluster 0 should be the top priority for aid like investing in health infrastructure since those countries face the biggest challenges. NGOs and other groups can use these insights to focus their resources where they're needed most.

Looking ahead, it makes sense to bring in time-series data to track trends over time, add supervised models (like logistic regression to predict who really needs aid), double-check results with hierarchical clustering, and bring in the latest data. That way, the picture just gets sharper.

VI. REFERENCES

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